Can a Frozen Pixel-Based World Model Decode the Position of an Occluded Object?

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Abstract

World models learn predictive representations of physical environments, but it remains unclear whether their internal states preserve the position of objects that are no longer visible. We study this prerequisite for belief representation using MineOcclude, a controlled Minecraft benchmark in which a sheep moves behind an occluding wall. Using 25 identity-grouped trajectories, we screen current-frame activations from 11 decoder layers of the frozen MineWorld model with global and coarse-spatial linear probes, then refine the advancing layer using training-only token selection. The strongest hidden-position candidate, layer-19 coarse 2×6 pooling, obtains macro $R^2 = -13.826$ and RMSE $= 2.071$ blocks, failing prespecified positive-R^2 and RMSE ≤ 0.50 requirements. In contrast, the same representation accurately decodes visible position both without a wall ($R^2 = 0.958$) and in the wall environment ($R^2 = 0.930$). Thus, no candidate in the evaluated current-frame linear-readout family qualifies for hidden-position decoding, despite a validated measurement pipeline and strong matched visible controls.

1 Introduction

A world model compresses the state-transition processes of a physical environment under finite computational resources. Effective world models should therefore represent objects that are temporarily unobserved and update those representations as time passes and evidence returns. Prior work shows that learned representations can encode environmental state and structured belief information [1–3], while recent transformer-based world models improve visual sequence modeling [4]. Nevertheless, recovering an object while it is visible does not establish that a pixel-based model preserves its state through occlusion. Even successful hidden-position decoding would be only a prerequisite for stronger claims about calibrated uncertainty or causal use.

We investigate this prerequisite using MineOcclude, a Minecraft benchmark in which a sheep follows controlled stochastic motion behind a wall. Synchronized server state supplies the true wall-parallel position, avoiding image-based target estimation. Paired no-wall and wall trajectories separate ordinary visible decoding, visible decoding in the wall domain, and decoding during complete occlusion. We probe the frozen MineWorld model rather than training it on the benchmark.

Our central experiment asks whether the negative layer-4 result from a preliminary study can be explained by choosing the wrong depth or by averaging away spatial information. We conduct a prespecified search over 11 decoder layers, global and coarse-spatial summaries, and a layer-19 token-selection refinement. Candidate discovery uses only fully occluded examples, while matched visible populations serve exclusively as controls. MineOcclude ultimately supports a broader progression from decodability to calibration, temporal belief updates, and intervention, but the present study evaluates only the first criterion; uncertainty calibration and causal use remain untested.

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2 Related Work

Research on world models shows that sequence models can encode latent state. Othello-GPT recovers board state from next-move prediction, and interventions alter its outputs [1]; linear probes and vector arithmetic expose manipulable features [2]. Causal probing recovers scene structure in latent diffusion models [5] and game variables in learned Atari simulators [6]. Transformer and diffusion world models for visual control likewise model latent dynamics and future frames [7, 8, 4]. These studies motivate probing, but mostly evaluate point states rather than what remains accessible when observations disappear.

A second axis concerns belief states under partial observability. POMDP theory identifies the posterior over latent states as the sufficient statistic for optimal decisions [9], and transformers trained on synthetic processes can represent belief-state geometry [3]. Pixel- and action-conditioned world models extend this setting to visual control, including MineWorld [10], but existing analyses do not establish whether a physically moving, fully occluded object remains linearly accessible. A point decoder can also succeed without representing spread, multimodality, or elapsed-time dependence, so decodability must be separated from calibration and causal use.

A third axis studies physical reasoning and object permanence in video models. Self-supervised video models distinguish physically plausible events [11]; video diffusion internals encode physical plausibility [12], and analyses localize emergent physics features [13]. Concept interventions can alter object-permanence judgments [14], while WorldBench evaluates related physical concepts behaviorally [15] and CoMotion tracks objects through occlusion with supervised state estimation [16]. Calibration work characterizes confidence in classifiers [17] and video models [18]. MineOcclude complements these approaches by holding the world model fixed, pairing internal probes with synchronized physical targets, and imposing prospective controls and decision gates before advancing to stronger belief claims.

3 Methodology

3.1 Benchmark and Paired Experimental Design

Experiment II reused 25 qualified trajectory identities from the paired MineOcclude apparatus. Each identity fixed the direction, stochastic motion schedule, and target lineage across wall and no-wall captures in `layout_01` at speed 2.0 blocks/s. The target was current, direction-specific, causal-FIR-aligned sheep world-x in blocks. Three populations isolated the relevant conditions: `NO_WALL` contained 475 visible rows, `WALL_VISIBLE` contained 846 visible rows outside complete occlusion, and `WALL_HIDDEN` contained 475 fully occluded rows. The trajectory identity was the independent unit; all rows from an identity remained together in one of five grouped outer folds.

3.2 Model Representations and Probes

We froze the MineWorld `700M_16f` checkpoint and processed complete 16-pair teacher-forced contexts. At the current pair, we captured the 336 visual-token vectors of width 2,048 at decoder layers $\{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 19\}$. The M5 screen evaluated two current-frame representations at every layer: GLOBAL MEAN averaged all token vectors, whereas COARSE 2X6 averaged each 7×4 cell of a 2×6 partition and concatenated the 12 resulting vectors. M6 retained these representations at the advancing layer and added BEST-K, whose tokens and $K \in \{1, 4, 16, 64\}$ were selected using training identities only. Ridge probes used train-only standardization, an unpenalized intercept, and $\alpha \in \{0.1, 1, 10, 100, 1000\}$. Inner grouped rotations selected hyperparameters; outer held-out identities supplied predictions.

3.3 Evaluation and Decision Criteria

M5 ranked 22 layer–representation cells using only `WALL_HIDDEN`; visible controls could not affect layer, representation, token, or hyperparameter selection. M6 refined three cells at the advancing layer, after which a frozen selector chose one finalist without runner-up substitution. M7 applied an identity bootstrap, both visible controls, and 49 full-funnel randomizations within direction-by-phase strata. Qualification required all nine prespecified gates, including positive improvement over

Table 1: Finalist performance across visibility conditions. Metrics are trajectory-macro averages.

Condition	R^2	RMSE	MAE
NO_WALL	0.9582	0.1090	0.0896
WALL_VISIBLE	0.9301	0.1051	0.0779
WALL_HIDDEN	-13.8263	2.0711	1.9244

baseline and a positive lower bound, macro $R^2 > 0$ with lower bound above zero, RMSE ≤ 0.50 blocks, fold/direction/phase stability, passing controls, and selection-aware $p_A \leq 0.05$. Metrics were macro-averaged over trajectory identities.

4 Experiments

4.1 Experimental Setup and Qualification

Outcome-blind qualification locked the current-frame tier before scientific results were available. Extraction then completed 1,796 model forwards and published 19,756 finite float16 activation tensors across the 11 sampled layers. Source, checkpoint, cohort, target, fold, tensor, and feature identities were frozen and verified. No capture, row, or candidate was replaced after inspection of outcomes.

4.2 Layer and Representation Search

M5 screened 22 current-frame cells. Coarse 2×6 pooling outperformed global pooling at every sampled layer, and layer 19 advanced. The leading cell, `L19|COARSE_2X6_CONCAT|CURRENT`, produced hidden macro $R^2 = -13.8263$, RMSE $= 2.0711$ blocks, and mean baseline improvement $\Delta_A = 0.3992$ blocks. M6 added training-only BEST-K at layer 19, but it performed worse ($R^2 = -14.9922$, RMSE $= 2.1453$); the coarse representation remained the finalist.

4.3 Final Qualification and Controls

The finalist accurately decoded position in both visible populations, demonstrating that the measurement path worked and that the wall domain alone did not explain failure. During complete occlusion, however, its bootstrap lower R^2 bound was -15.4806 and its RMSE exceeded the 0.50-block ceiling by 1.5711 blocks. Its improvement interval, $[0.3485, 0.4537]$, and selection-aware $p_A = 0.02$ indicated a weak structured difference from the specified null, but not scientifically adequate reconstruction: the null mean was 0.3846 and its maximum lay only 0.0002 below the observed statistic. The finalist failed the positive-R^2 and RMSE gates, so zero candidates qualified.

5 Discussion

Experiment II addresses two major explanations left open by the preliminary layer-4 result. Later depth and coarse spatial preservation affected the ranking: layer 19 was best, and coarse pooling consistently exceeded global pooling. Their practical effect was nevertheless small. Hidden RMSE improved by only about 0.039 blocks relative to the earlier result and remained more than four times the qualification ceiling; token selection worsened performance.

The strong `NO_WALL` and `WALL_VISIBLE` controls make a broken extraction/probe pipeline and generic wall-domain shift unlikely explanations. At the same time, $p_A = 0.02$ should not be confused with predictive adequacy. The selected statistic only narrowly exceeded all 49 null reruns, while absolute prediction was substantially worse than the mean-prediction baseline. The evidence therefore supports a weak subthreshold structure, not a qualified hidden-position code.

This negative is deliberately bounded. The study sampled 11 of 20 decoder layers in one layout and speed, used current-frame linear ridge probes, and did not evaluate temporal differences, multi-frame concatenation, nonlinear probes, full-grid/PCA/wall-region representations, other layouts, or other speeds. It also did not test posterior uncertainty, calibration, or causal use. Consequently, "no qualifying decoder in the evaluated family" must not be interpreted as model-wide absence of hidden information.

6 Conclusion

MineOcclude provides a controlled test of whether a frozen pixel-based world model preserves the position of a fully occluded object. Across 11 sampled decoder layers and prespecified current-frame linear readouts, no candidate met the complete qualification criteria. The same selected representation decoded visible position accurately with and without the wall, localizing the failure to complete occlusion in this apparatus rather than to the measurement pipeline. Depth and coarse spatial structure exposed only a weak, scientifically inadequate signal. Because hidden-position decodability did not qualify, stronger claims about calibrated uncertainty, temporal belief dynamics, and causal intervention remain untested.

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